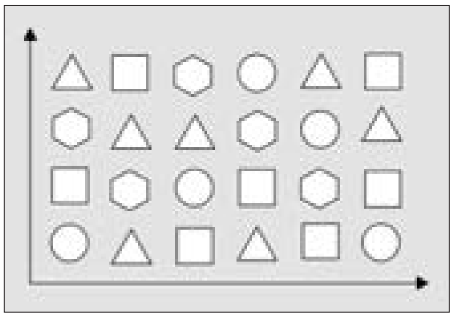
Code Division Multiple Access - CDMA

While in FDMA and TDMA different calls are assigned different frequencies, in CDMA, all callers always occupy the same frequency band. Using a technique called *Direct Sequence Spread Spectrum*, the caller's voice is 'spread' over the entire frequency band by multiplying it with a unique high-frequency *pseudo-random code*, which for all practical purposes can be considered completely random. The result is a signal that is spread out both in time and frequency. The pseudo-random code is generated by the MTSO and shared only with the two mobile phones on a call. The code is used again at the receiving end to recover the original voice signal. These codes are so distinct that the possibility of one call interfering with another is reduced to a minimum. Theoretically, the number of possible codes is infinite; so on paper, a CDMA network can handle an unlimited number of subscribers.



Our conversations spread out over all frequencies in a CDMA network

cdmaOne is a standard set for CDMA based networks, proposed by Qualcomm and approved by the TIA as Interim Standard 95 (IS-95). cdmaOne networks use advanced voice compression techniques to improve the efficiency of the system, such as a variable bit-rate *Vocoder (Voice Encoder)*. A vocoder basically converts our analogue voice signal to a digital signal. Vocoders were not a new concept - they were used in all digital communication. However, these were fixed bit-rate vocoders. This meant that they would always transfer data at, say, 2400 bits per second - even when there is no talking going on. The variable bit-rate vocoder would idle at something like 800 bits per second when nobody was talking, and would